

BCS Limit and BEC Limit

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1 BCS Limit

1.1 s-wave case

Consider the gap equation after renormalization. When $T = T_c$, then $\Delta_0 = 0$ and $E_{\mathbf{k}} = \xi_{\mathbf{k}}$. Here, we set $\hbar = k_B = 1$ and $\beta_c = 1/T_c$, thus

$$\begin{aligned} -\frac{m}{4\pi a_s} &= \frac{1}{V} \sum_{\mathbf{k}} \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) \\ &= \frac{1}{2\pi^2} \int_0^\infty dk \, k^2 \left[\frac{\tanh \frac{\beta_c (\epsilon_{\mathbf{k}} - \mu)}{2}}{2(\epsilon_{\mathbf{k}} - \mu)} - \frac{1}{2\epsilon_{\mathbf{k}}} \right] \\ &= \frac{1}{4\pi^2} \cdot (2m)^{\frac{3}{2}} \int_0^\infty d\epsilon \, \sqrt{\epsilon} \left[\frac{\tanh \frac{\beta_c (\epsilon - \mu)}{2}}{2(\epsilon - \mu)} - \frac{1}{2\epsilon} \right] \\ &= \frac{mk_F}{2\pi^2} \int_0^\infty dx \, \sqrt{x} \left\{ \frac{\tanh [y(x-1)]}{2(x-1)} - \frac{1}{2x} \right\}, \end{aligned} \tag{1}$$

where $x = \epsilon/\mu$ and $y = \beta_c \mu/2$. In weak coupling, $\mu \simeq \epsilon_F \gg T_c$, i.e., y is large, and we have

$$\tanh [y(x-1)] = 1 - 2\theta(1-x). \tag{2}$$

To further evaluate the integral from Eq.(1), we first use that

$$\frac{\sqrt{x}}{x-1} = \frac{1}{\sqrt{x}+1} + \frac{1}{x-1}, \quad (3)$$

such that we have two integrals to perform. The first integral is given by

$$\begin{aligned} \frac{mk_F}{2\pi^2} \int_0^{x_\Lambda} dx \left\{ \frac{\tanh[y(x-1)]}{2(\sqrt{x}+1)} - \frac{1}{2\sqrt{x}} \right\} &\simeq \frac{mk_F}{4\pi^2} \left[\int_0^1 dx \left(\frac{-1}{\sqrt{x}+1} - \frac{1}{\sqrt{x}} \right) + \int_1^{x_\Lambda} dx \left(\frac{1}{\sqrt{x}+1} - \frac{1}{\sqrt{x}} \right) \right] \\ &= \frac{mk_F}{4\pi^2} [-4 + 2\ln 4 - 2\ln(1 + \sqrt{x_\Lambda})]. \end{aligned} \quad (4)$$

where we take the limit $x_\Lambda \rightarrow \infty$ at the end of the calculation. And the second integral is given by

$$\frac{mk_F}{2\pi^2} \int_0^{x_\Lambda} dx \frac{\tanh[y(x-1)]}{2(x-1)} \xrightarrow{\tilde{x}=x-1} \frac{mk_F}{4\pi^2} \int_{-1}^{x_\Lambda-1} d\tilde{x} \frac{\tanh(y\tilde{x})}{\tilde{x}} \quad (5)$$

which we can further evaluate by using

$$\begin{aligned} \int_{-1}^{x_\Lambda-1} d\tilde{x} \frac{\tanh(y\tilde{x})}{\tilde{x}} &= \int_{-1}^0 d\tilde{x} \frac{\tanh(y\tilde{x})}{\tilde{x}} + \int_0^{x_\Lambda-1} d\tilde{x} \frac{\tanh(y\tilde{x})}{\tilde{x}} \\ &\xrightarrow{z=-\tilde{x}} \int_1^0 (-dz) \frac{\tanh(-yz)}{-z} + \int_0^{x_\Lambda-1} d\tilde{x} \frac{\tanh(y\tilde{x})}{\tilde{x}} \\ &= \int_0^1 dz \frac{\tanh(yz)}{z} + \int_0^{x_\Lambda-1} dz \frac{\tanh(yz)}{z} \\ &\xrightarrow{u=yz} \int_0^y du \frac{\tanh u}{u} + \int_0^{y(x_\Lambda-1)} du \frac{\tanh u}{u} \\ &= \tanh u \cdot \ln u \Big|_0^y - \int_0^y du \frac{\ln u}{\cosh^2 u} + \tanh u \cdot \ln u \Big|_0^{y(x_\Lambda-1)} - \int_0^{y(x_\Lambda-1)} du \frac{\ln u}{\cosh^2 u} \\ &\simeq \ln y + \ln[y(x_\Lambda-1)] - 2 \int_0^\infty du \frac{\ln u}{\cosh^2 u} \\ &= \ln(x_\Lambda-1) + 2\ln y - 2\ln \frac{\pi}{4e^\gamma}, \end{aligned} \quad (6)$$

where $\gamma = 0.5772$ is Euler's constant. In the last step, we have used the relation

$$\begin{aligned} \int_0^\infty dx \frac{\ln x}{\cosh^2 x} &= \psi\left(\frac{1}{2}\right) + \ln \pi \\ &= -\gamma - 2\ln 2 + \ln \pi \\ &= \ln \frac{\pi}{4e^\gamma}, \end{aligned} \quad (7)$$

which can be found in the textbook *Table of Integrals, Series and Products*, or computed directly using Wolfram Mathematica.

Above all, we obtain that

$$\begin{aligned}
-\frac{m}{4\pi a_s} &= \frac{mk_F}{4\pi^2} \left[-4 + 2\ln 4 - 2\ln(1 + \sqrt{x_\Lambda}) + \ln(x_\Lambda - 1) + 2\ln y - 2\ln \frac{\pi}{4e^\gamma} \right] \\
&= \frac{mk_F}{4\pi^2} \left(-4 + 2\ln 4 + \ln \frac{x_\Lambda - 1}{(1 + \sqrt{x_\Lambda})^2} + 2\ln y - 2\ln \frac{\pi}{4e^\gamma} \right) \\
&= \frac{mk_F}{4\pi^2} \left(-4 + 2\ln 4 + 2\ln \frac{\epsilon_F}{2T_c} - 2\ln \frac{\pi}{4e^\gamma} \right),
\end{aligned} \tag{8}$$

where we finally took the limit $x_\Lambda \rightarrow \infty$. Then, Eq.(8) can be solved for T_c , giving in the case of a small negative scattering length a_s the analytic result

$$T_c = \frac{8}{\pi} \epsilon_F \exp \left(\gamma - 2 - \frac{\pi}{2k_F |a_s|} \right). \tag{9}$$

1.2 p-wave case

Also, consider the gap equation after renormalization. When $T = T_c$, then $\Delta_0 = 0$ and $E_{\mathbf{k}} = \xi_{\mathbf{k}}$

$$-\frac{m}{16\pi^2 a_p k_0^2} = \frac{1}{V} \sum_{\mathbf{k}} |\Gamma(\mathbf{k})|^2 \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right), \tag{10}$$

where $\Gamma(\mathbf{k})$ takes the form

$$\begin{aligned}
\Gamma(\mathbf{k}) &= \Gamma_k(k) \Gamma_a(\hat{\mathbf{k}}) \\
&= \frac{k k_0}{k^2 + k_0^2} Y_{1,0}(\hat{\mathbf{k}}),
\end{aligned} \tag{11}$$

and $Y_{1,0}(\hat{\mathbf{k}})$ is the spherical harmonics for the angular dependence. Moreover, $k_0 \sim R_0^{-1}$ sets the momentum scale, where R_0 is the interaction range in real space. For diluteness condition $nR_0^3 \ll 1$ requires $k_F/k_0 \ll 1$. In BCS limit, the gap equation now takes the form

$$\begin{aligned}
-\frac{m}{16\pi^2 a_p k_0^2} &= \int \frac{d^3 k}{(2\pi)^3} |\Gamma(\mathbf{k})|^2 \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) \\
&= \frac{1}{8\pi^3} \int_0^\infty dk \, k^2 \Gamma_k^2(k) \left(\frac{\tanh \frac{\beta_c \xi_k}{2}}{2\xi_k} - \frac{1}{2\epsilon_k} \right) \\
&= \frac{1}{8\pi^3} \int_0^\infty dk \, k^2 [\Gamma_k^2(k_F) + \Gamma_k^2(k) - \Gamma_k^2(k_F)] \left(\frac{\tanh \frac{\beta_c \xi_k}{2}}{2\xi_k} - \frac{1}{2\epsilon_k} \right),
\end{aligned} \tag{12}$$

where it is divided into two parts.

The contribution from the first part is given by

$$\begin{aligned}
\frac{1}{8\pi^3} \int_0^\infty dk \ k^2 \Gamma_k^2(k_F) \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) &\simeq \frac{1}{8\pi^3} \cdot \frac{k_F^2}{k_0^2} \int_0^\infty dk \ k^2 \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) \\
&= \frac{1}{8\pi^3} \cdot \frac{k_F^2}{k_0^2} \cdot m k_F \left(-2 + \ln 4 + \ln \frac{\epsilon_F}{2T_c} - \ln \frac{\pi}{4e^\gamma} \right) \\
&= \frac{m}{8\pi^3} \cdot \frac{k_F^3}{k_0^2} \left(-2 + \ln 4 + \ln \frac{\epsilon_F}{2T_c} - \ln \frac{\pi}{4e^\gamma} \right), \tag{13}
\end{aligned}$$

which calculation process is similar to the s -wave case. And the contribution from the other part is given by

$$\begin{aligned}
\frac{1}{8\pi^3} \int_0^\infty dk \ k^2 [\Gamma_k(k) - \Gamma_k(k_F)] \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) &= \frac{1}{8\pi^3} \int_0^\infty dk \ k^2 \left[\frac{k^2 k_0^2}{(k^2 + k_0^2)^2} - \frac{k_F^2 k_0^2}{(k_F^2 + k_0^2)^2} \right] \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) \\
&= \frac{1}{8\pi^3} \int_0^\infty dk \ k^2 \frac{k_0^2 (k^2 - k_F^2) (k_0^4 - k^2 k_F^2)}{(k^2 + k_0^2)^2 (k_F^2 + k_0^2)^2} \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) \\
&= \frac{m}{8\pi^3} \int_0^\infty dk \ k^2 \frac{k_0^2 (k_0^4 - k^2 k_F^2)}{(k^2 + k_0^2)^2 (k_F^2 + k_0^2)^2} \left(\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2} - \frac{k^2 - k_F^2}{k^2} \right). \tag{14}
\end{aligned}$$

Using Eq.(2), we can further separate Eq.(14) into two parts, as follows

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$$\begin{aligned}
\frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_0^{k_F} dk \ k^2 \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} \left(-1 - \frac{k^2 - k_F^2}{k^2} \right) &= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_0^{k_F} dk \ \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} [-k^2 - (k^2 - k_F^2)] \\
&= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_0^{k_F} dk \ \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} (-2k^2 + k_F^2), \tag{15}
\end{aligned}$$

•

$$\begin{aligned}
\frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_{k_F}^\infty dk \ k^2 \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} \left(1 - \frac{k^2 - k_F^2}{k^2} \right) &= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_{k_F}^\infty dk \ \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} [k^2 - (k^2 - k_F^2)] \\
&= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_{k_F}^\infty dk \ \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} k_F^2. \tag{16}
\end{aligned}$$

After simplification, we obtain

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$$\begin{aligned}
\frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_0^\infty dk \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} k_F^2 &= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \cdot k_F^2 \left[\int_0^\infty dk \frac{k_0^4}{(k^2 + k_0^2)^2} - \int_0^\infty dk \frac{k^2 k_F^2}{(k^2 + k_0^2)^2} \right] \\
&= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \cdot k_F^2 \left(\frac{\pi}{4} k_0 - \frac{\pi}{4} \frac{k_F^2}{k_0} \right) \\
&= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \cdot k_F^2 \cdot \frac{\pi}{4 k_0} (k_0^2 - k_F^2) \\
&= \frac{m}{8\pi^3} \cdot \frac{\frac{1}{k_0^2}}{\left[\left(\frac{k_F}{k_0} \right)^2 + 1 \right]^2} \cdot k_F^2 \cdot \frac{\pi}{4} k_0 \left[1 - \left(\frac{k_F}{k_0} \right)^2 \right] \\
&\simeq \frac{m}{8\pi^3} \cdot \frac{1}{k_0^2} \cdot k_F^2 \cdot \frac{\pi}{4} k_0 \\
&= \frac{m}{8\pi^3} \cdot \frac{\pi k_F^2}{4 k_0},
\end{aligned} \tag{17}$$

•

$$\begin{aligned}
\frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \int_0^{k_F} dk \frac{k_0^4 - k^2 k_F^2}{(k^2 + k_0^2)^2} (-2k^2) &= \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \cdot (-2) \left[\int_0^{k_F} dk \frac{k^2 k_0^4}{(k^2 + k_0^2)^2} - \int_0^{k_F} dk \frac{k^4 k_F^2}{(k^2 + k_0^2)^2} \right] \\
&\simeq \frac{m}{8\pi^3} \cdot \frac{k_0^2}{(k_F^2 + k_0^2)^2} \cdot (-2) \cdot \frac{k_F^3}{3} \\
&\simeq \frac{m}{8\pi^3} \cdot \frac{1}{k_0^2} \cdot (-2) \cdot \frac{k_F^3}{3} \\
&= -\frac{m}{8\pi^3} \cdot \frac{2k_F^3}{3k_0^2}.
\end{aligned} \tag{18}$$

Hence, the result of Eq.(14) is given by

$$\frac{1}{8\pi^3} \int_0^\infty dk \ k^2 [\Gamma_k(k) - \Gamma_k(k_F)] \left(\frac{\tanh \frac{\beta_c \xi_{\mathbf{k}}}{2}}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) = \frac{m}{8\pi^3} \cdot \frac{\pi k_F^2}{4 k_0} - \frac{m}{8\pi^3} \cdot \frac{2k_F^3}{3k_0^2} \tag{19}$$

The detail calculation of Eq.(18) reads

◦

$$\begin{aligned}
\int_0^{k_F} dk \frac{k^2 k_0^4}{(k^2 + k_0^2)^2} &= k_0^4 \left(\frac{1}{2k_0} \arctan \frac{k_F}{k_0} - \frac{1}{2} \frac{k_F}{k_F^2 + k_0^2} \right) \\
&\simeq k_0^4 \left\{ \frac{1}{2k_0} \left[\frac{k_F}{k_0} - \frac{1}{3} \left(\frac{k_F}{k_0} \right)^3 \right] - \frac{1}{2} \frac{k_F}{k_0^2 \left[\left(\frac{k_F}{k_0} \right)^2 + 1 \right]} \right\} \\
&\simeq k_0^4 \left\{ \frac{1}{2k_0} \left[\frac{k_F}{k_0} - \frac{1}{3} \left(\frac{k_F}{k_0} \right)^3 \right] - \frac{1}{2} \frac{k_F}{k_0^2} \left[1 - \left(\frac{k_F}{k_0} \right)^2 \right] \right\} \\
&= k_0^4 \left(\frac{1}{2} \frac{k_F}{k_0^2} - \frac{1}{6} \frac{k_F^3}{k_0^4} - \frac{1}{2} \frac{k_F}{k_0^2} + \frac{1}{2} \frac{k_F^3}{k_0^4} \right) \\
&= \frac{k_F^3}{3},
\end{aligned} \tag{20}$$

◦

$$\begin{aligned}
\int_0^{k_F} dk \frac{k^4 k_F^2}{(k^2 + k_0^2)^2} &= k_F^2 \left(k_F - \frac{3}{2} k_0 \arctan \frac{k_F}{k_0} + \frac{1}{2} \frac{k_0^2 k_F}{k_F^2 + k_0^2} \right) \\
&\simeq k_F^2 \left\{ k_F - \frac{3}{2} k_0 \left[\frac{k_F}{k_0} - \frac{1}{3} \left(\frac{k_F}{k_0} \right)^3 \right] + \frac{1}{2} \frac{k_0^2 k_F}{k_0^2 \left[\left(\frac{k_F}{k_0} \right)^2 + 1 \right]} \right\} \\
&\simeq k_F^2 \left\{ k_F - \frac{3}{2} k_0 \left[\frac{k_F}{k_0} - \frac{1}{3} \left(\frac{k_F}{k_0} \right)^3 \right] + \frac{1}{2} k_F \left[1 - \left(\frac{k_F}{k_0} \right)^2 \right] \right\} \\
&= k_F^2 \left(k_F - \frac{3}{2} k_F + \frac{1}{2} \frac{k_F^3}{k_0^2} + \frac{1}{2} k_F - \frac{1}{2} \frac{k_F^3}{k_0^2} \right) \\
&= 0.
\end{aligned} \tag{21}$$

Finally, the above results yield

$$-\frac{m}{16\pi^2 a_p k_0^2} = \frac{m}{8\pi^3} \cdot \frac{k_F^3}{k_0^2} \left(-2 + \ln 4 + \ln \frac{\epsilon_F}{2T_c} - \ln \frac{\pi}{4e\gamma} \right) + \frac{m}{8\pi^3} \cdot \frac{\pi k_F^2}{4k_0} - \frac{m}{8\pi^3} \cdot \frac{2k_F^3}{3k_0^2}, \tag{22}$$

which can be solved for T_c

$$T_c = \frac{8}{\pi} \epsilon_F \exp \left(\gamma - \frac{8}{3} + \frac{\pi k_0}{4k_F} - \frac{\pi}{2k_F^3 |a_p|} \right). \tag{23}$$

2 BEC Limit

2.1 s-wave case

In BEC limit, μ is negative and large, let $|\mu| = -\mu$. For Eq.(1) now we have

$$\begin{aligned}
-\frac{m}{4\pi a_s} &\simeq \frac{1}{2\pi^2} \int_0^\infty dk \, k^2 \left(\frac{1}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) \\
&= \frac{1}{2\pi^2} \int_0^\infty dk \, k^2 \left(\frac{m}{k^2 + 2m|\mu|} - \frac{m}{k^2} \right) \\
&= \frac{1}{2\pi^2} \int_0^\infty dk \, \frac{-2m^2|\mu|}{k^2 + 2m|\mu|} \\
&= \frac{1}{2\pi^2} (-2m^2|\mu|) \frac{\pi}{2\sqrt{2m|\mu|}},
\end{aligned} \tag{24}$$

such that

$$\mu = -\frac{1}{2ma_s^2}. \tag{25}$$

2.2 p-wave case

In BEC limit, μ is negative and large, let $|\mu| = -\mu$. For Eq.(10) now we have

$$\begin{aligned}
-\frac{m}{16\pi^2 a_p k_0^2} &\simeq \frac{1}{8\pi^3} \int_0^\infty dk \, k^2 \frac{k^2 k_0^2}{(k^2 + k_0^2)^2} \left(\frac{1}{2\xi_{\mathbf{k}}} - \frac{1}{2\epsilon_{\mathbf{k}}} \right) \\
&= \frac{1}{8\pi^3} \int_0^\infty dk \, k^2 \frac{k^2 k_0^2}{(k^2 + k_0^2)^2} \left(\frac{m}{k^2 + 2m|\mu|} - \frac{m}{k^2} \right) \\
&= \frac{1}{8\pi^3} \int_0^\infty dk \, \frac{k^2 k_0^2}{(k^2 + k_0^2)^2} \frac{-2m^2|\mu|}{k^2 + 2m|\mu|} \\
&= \frac{1}{8\pi^3} (-2m^2|\mu| k_0^2) \frac{\pi}{4k_0 \left(k_0 + \sqrt{2m|\mu|} \right)^2},
\end{aligned} \tag{26}$$

such that

$$\mu = -\frac{1}{mk_0 a_p}, \tag{27}$$

where we have used the condition $a_p > 2/k_0^3$.